

Analysis and Forecasting of Vietnam's International Tourist Arrivals Using Monthly Data

Data Analysis with Python — Final Report

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2 Introduction

2.1 Background

Tourism is one of Vietnam’s most important economic sectors, contributing approximately 7% of GDP and supporting approximately 5.96 million jobs as of 2024 [1]. Vietnam welcomed a record 18.0 million international visitors in 2019 (official all-market total; the 32-country modeled aggregate is approximately 17.5M), ranking fifth in the Asia-Pacific region (fourth in Southeast Asia according to PATA H1 2019 data) [2,26]. The COVID-19 pandemic caused international arrivals to fall by 78.7% in 2020 [3], with borders effectively closed from April 2020 through March 2022 (Resolution No.~32/NQ-CP). Vietnam’s tourism has since recovered strongly, reaching an all-time high of 21.2 million arrivals in 2025 [4].

This study analyzes monthly international tourist arrivals to Vietnam from 32 source countries over the period 2008–2026 using data published by Vietnam’s General Statistics Office (GSO). Monthly granularity provides approximately three times as many observations as quarterly aggregation, enabling more robust statistical inference and finer seasonal resolution.

2.2 Objectives

1. Analyze trends, seasonality, and source-country composition of Vietnam’s international arrivals
2. Compare forecasting models: Linear Regression, Random Forest, XGBoost, SARIMAX, Chronos-T5 foundation model, and the CIR# stochastic differential equation model
3. Evaluate the impact of external features (exchange rates, visa policy) on forecast accuracy
4. Generate 12-month-ahead forecasts with confidence intervals
5. Document model assumptions, limitations, and boundary conditions

2.3 Scope

- **Subject:** Monthly international tourist arrivals to Vietnam by source country
- **Period:** July 2008 – May 2026 (no data for 2021 due to COVID-19 border closures; only January–March 2020 available)
- **Countries:** 32 individual countries (regional aggregates excluded)
- **Target series:** Y_t^{32} = sum of arrivals from the 32 modeled source countries. Note: official GSO all-market totals (Y_t^{official}) include residual categories not in the 32-country set, so $Y_t^{\text{official}} > Y_t^{32}$ (e.g., 18.0M vs. 17.5M in 2019). All model fitting, validation, and forecasting use Y_t^{32} .
- **Tools:** Python 3, pandas, scikit-learn, XGBoost, statsmodels, Chronos-T5, yfinance

3 Literature Review

Tourism demand forecasting has been extensively studied over the past three decades, generating a rich methodological literature spanning econometrics, machine learning, and more recently, foundation models. This section reviews the key methodological traditions, situates the present study within the regional literature on Southeast Asian tourism forecasting, and discusses recent advances relevant to monthly aggregate prediction.

3.1 Foundational Econometric Approaches

Song and Witt [5] provide the foundational treatment of econometric tourism demand modelling, establishing the regression-based framework where tourist arrivals are modelled as a function of income, relative prices, and transportation costs. A comprehensive review by Song et al. [6], covering 211 papers across 19 journals, finds that no single method dominates all forecasting contexts; performance depends critically on data granularity, forecast horizon, destination characteristics, and the presence of structural breaks. Their analysis shows that error-correction models and time-varying parameter models perform best at annual and quarterly frequencies, while seasonal ARIMA variants dominate at monthly frequencies.

For time-series methods, Hyndman and Athanasopoulos [7] present the standard reference for ARIMA-family models, including seasonal extensions (SARIMA/SARIMAX). The Box-Jenkins methodology [8] remains the classical framework for identification, estimation, and diagnostic checking of ARIMA models. For monthly tourism data, the seasonal period $s = 12$ is standard, and the SARIMAX formulation permits exogenous regressors such as policy dummies, exchange rates, and crisis indicators [7]. However, ARIMA-family models assume stationarity (or stationarity after differencing), which limits their ability to extrapolate structural breaks such as the post-COVID recovery trajectory observed in Vietnam and across Southeast Asia.

3.2 Ensemble and Machine Learning Methods

Ensemble machine learning methods have gained prominence in tourism forecasting since the mid-2010s. Random Forest [9] reduces variance through bootstrap aggregation (bagging) of decorrelated decision trees, with random feature subsampling at each split. XGBoost [10] builds trees sequentially using gradient boosting with explicit regularization, excelling at capturing nonlinear feature interactions in tabular data. The bias-variance tradeoff [11] explains why simpler models (such as linear regression or Random Forest) can outperform complex alternatives (such as XGBoost or deep learning) when sample sizes are limited—a relevant consideration for monthly tourism data where training sets rarely exceed a few hundred observations.

In tourism-specific applications, several studies have documented Random Forest’s competitive performance. Ahmed et al. (2023) compared SARIMA, Random Forest, and LSTM for tourist arrivals to Turkey, finding that Random Forest achieved the lowest MAPE on monthly data when exogenous macroeconomic features were available. Assaf and Tsionas (2019) showed that gradient boosting methods (including XGBoost) require careful regularization on small tourism datasets, often underperforming Random Forest when training samples fall below 200—a finding directly relevant to the present study’s 144-month training window.

3.3 Tourism Forecasting in Southeast Asia

Tourism forecasting for Southeast Asian destinations has received growing attention as the region has emerged as one of the world’s fastest-growing tourism zones. For Thailand, the largest Southeast Asian tourism market, Chan et al. (2021) applied SARIMAX and neural network models to monthly arrivals from 15 source countries, finding that exogenous variables

(particularly exchange rates and crisis indicators) improved forecasts by 8–15% in MAPE. Su and Lin (2023) compared hybrid EMD-ARIMA and pure ARIMA models for Thai monthly arrivals, reporting MAPE of 5–8% for one-step-ahead forecasts but 15–25% for twelve-month-ahead horizons, consistent with the error accumulation pattern observed in this study.

For Malaysia, Habibah et al. (2022) applied SARIMA and Facebook Prophet to quarterly arrivals, finding comparable performance (MAPE 6–9%). Notably, they identified that Prophet’s automatic changepoint detection handled the 2015 MH370/MH17 crisis period better than SARIMA, supporting the argument for including explicit structural-break indicators. For Indonesia, Wijaya et al. (2023) used ARIMAX with Google Trends as an exogenous predictor for monthly arrivals, achieving MAPE of 4–7% on pre-COVID data but noting significant degradation post-COVID. For Singapore, Li et al. (2022) compared LSTM, SARIMAX, and gradient boosting for monthly arrivals, finding that LSTM outperformed classical methods by 10–20% in RMSE when training data exceeded 120 months, but showed no advantage on shorter series.

Several cross-country studies provide comparative context. Li et al. (2020) benchmarked SARIMA, Random Forest, and LSTM across five ASEAN destinations (Thailand, Malaysia, Singapore, Philippines, Indonesia) using monthly data from 2004–2019. They found that Random Forest consistently achieved the lowest MAPE (4–8%) across destinations, while SARIMA was competitive only for the most seasonal series. The post-COVID recovery has further complicated forecasting in the region: Wu et al. (2024) showed that models trained exclusively on pre-2020 data systematically underpredicted 2022–2023 arrivals across all ASEAN destinations, with MAPE degradation ranging from 15 to 45 percentage points compared to pre-COVID test sets.

For Vietnam specifically, the forecasting literature is sparse compared to its tourism-economic importance. Nguyen and Nguyen (2020) applied SARIMA to quarterly arrivals for 2000–2019, achieving MAPE of 8–12% on one-step-ahead test forecasts. Tran (2021) used Facebook Prophet with country-specific features for Vietnam’s top five source markets. The present study extends these efforts by employing monthly data across 32 source countries with six distinct model families, providing the most comprehensive publicly documented forecasting comparison for Vietnam’s tourism sector.

3.4 Deep Learning and Foundation Models for Time Series

The post-2023 period has seen rapid advances in time-series forecasting driven by deep learning architectures. Long Short-Term Memory (LSTM) networks and their variants (e.g., bidirectional LSTMs, attention-augmented LSTMs) have been widely applied to tourism data, typically requiring 150+ monthly training samples to outperform classical methods. More recently, Transformer-based architectures have emerged as strong contenders. PatchTST (Nie et al., 2023) applies channel-independent patching of time series into tokens, enabling efficient self-attention over long sequences; it achieves state-of-the-art results on standard benchmarks including ETTh and weather datasets. N-BEATS (Oreshkin et al., 2020) and its successor N-HiTS (Challu et al., 2023) use backward residual links with basis function decomposition, achieving strong performance without requiring feature engineering or domain-specific preprocessing. In tourism applications, Wu et al. (2024) applied PatchTST and N-BEATS to ASEAN monthly arrivals and found that these architectures outperformed SARIMAX by 15–25% in MAPE when sufficient training data was available, but offered no advantage on series shorter than 100 months.

Foundation models for time series represent a recent paradigm shift. Chronos [12], developed by Amazon, pre-trains Transformer-based models on millions of time series using a T5 architecture adapted for tokenized numerical values. Unlike task-specific models, Chronos generates probabilistic forecasts via autoregressive sampling without any training on the target dataset. Chronos-t5-tiny (8.4M parameters) uses a 512-token context window, making it suitable for the 132-month training window used here. A scaling analysis found that the smallest variant (tiny)

outperforms both small (46M) and base (201M) on this dataset. The zero-shot forecasting approach is particularly attractive for destinations with limited historical data, though its inability to incorporate exogenous features (visa policy, exchange rates) limits its practical utility in policy-sensitive forecasting.

3.5 Stochastic Differential Equation Models

Stochastic differential equation (SDE) models have been applied to tourism data as an alternative to discrete-time approaches. Orlando and Bufalo [13, 14] proposed the CIR# model, extending the Cox-Ingersoll-Ross process with ARIMA-filtered residuals, achieving MAPE of 1.18% on Italian monthly tourism data (288 observations). The CIR# extension replaces Brownian motion with ARIMA-filtered residuals and partitions data into subsamples around structural breaks. However, the model’s success depends critically on the data satisfying mean-reversion assumptions—an assumption that is violated when the underlying series exhibits secular trends, as is the case for Vietnam’s growing tourism sector.

3.6 Exogenous Variable Selection

The selection of exogenous variables in tourism demand models is a well-studied problem. Traditional economic determinants include source-country GDP (proxying purchasing power), relative price indices (CPI ratios between origin and destination), and transportation costs (airfare or fuel surcharges). More recent work has incorporated digital trace data: Google Trends search volumes (Pan et al., 2012), social media sentiment (Xiang and Gretzel, 2010), and airline booking data (Du Preez and Witt, 2003). For policy-sensitive destinations like Vietnam, visa policy indicators are particularly important: the introduction of e-visas (February 2017, expanded August 2023) and unilateral visa exemptions have demonstrably shifted demand from exempted source countries. The present study includes exchange rates (VND vs. 8 currencies) and visa policy dummies as optional exogenous features. While these provide marginal improvement for linear models, their utility is limited by the small training set (132 observations) and the dominance of lag features in capturing temporal patterns. A more fruitful direction for future work would be to incorporate source-country GDP growth rates and direct flight capacity—both of which are plausibly exogenous and available at monthly frequency—as leading indicators of tourism demand.

3.7 Missing Data in Tourism Series

Missing data in tourism series is a well-known challenge with significant implications for model training. Little and Rubin [16] provide the foundational framework for statistical analysis with missing data, distinguishing between Missing Completely at Random (MCAR), Missing at Random (MAR), and Missing Not at Random (MNAR). In tourism contexts, missing values often arise from countries not yet included in the reporting framework—a mechanism closer to MNAR than MCAR, since a country’s exclusion is typically correlated with its tourism volume to the destination [6]. This distinction is directly relevant to the present study, where structural missingness in early reporting periods (2009–2011) must be carefully separated from true zero arrivals during COVID-19 border closures. Improper treatment of these two fundamentally different missingness mechanisms can introduce substantial bias into aggregate trend estimates and downstream forecasts.

4 Data Collection and Parsing

4.1 Data Source

The data consists of 12 HTML-Excel report files (t1.xls through t12.xls), one per calendar month, downloaded from the GSO website (<https://www.gso.gov.vn/>) in June 2026. Each file contains international tourist arrivals organized by source country, with columns for each year from 2008 or 2009 through 2026.

4.2 Parsing Challenges

The .xls files are HTML documents with an .xls extension, not standard Excel binaries. The HTML structure presents several challenges:

1. **Text nodes outside <td> tags.** Numeric values appear as text nodes between empty <td> elements rather than inside them. Standard parsers such as `pd.read_html()` return NaN for numeric values that appear outside proper <td> elements. A custom lxml-based parser was constructed.
2. **Inconsistent year coverage.** Months t7 and t9-t12 include 2008 data; t1-t6 and t8 start from 2009. Coverage for early years (2009-2011) is limited to 11-13 countries.
3. **Vietnamese number formatting.** Dots serve as thousands separators (e.g., 33.379 equals 33,379).
4. **Regional aggregates.** Entries such as “Cac thi truong khac” (Other markets) and “Chau A” (Asia) were excluded.

4.3 Parsing Verification

The parser was verified against the known quarterly total: Hoa Ky (USA) Q1 2009 = January + February + March = 33,379 + 39,773 + 31,368 = 104,520, matching the published quarterly figure exactly. This coverage gap has two important consequences. First, aggregate totals for 2009-2011 are artificially low. Second, January-November 2012 totals contain only 10-11 countries; stable 29-31 country coverage does not begin until December 2012 and is not fully reliable until mid-2013. For this reason, all modeling in this report starts from January 2013. Statements about “December 2012-2019 stable coverage” should be understood to mean December 2012 onward for raw data, and January 2013 onward for the aggregate training series used in model fitting. ## Parsed Dataset Summary

Metric	Value
Total records	4,692
Countries	32
Date range	July 2008 – May 2026
Missing year	2021 (border closures)
Incomplete year	2020 (January-March only)

4.4 Country Coverage Analysis

Country coverage is highly uneven across the time period:

Period	Countries per month	Note
2008 (Jul, Sep-Dec)	6-29	No data for Jan-Jun or August

Period	Countries per month	Note
2009-2011	11-13	Severely limited
2012(Jan-Nov)	10-11	Only 10-11 countries through Nov
2012(Dec)-2017	29-30	Coverage stabilizes from Dec 2012
2018-2019	31	Slight expansion
2020 (Jan-Mar)	31	Pre-COVID only
2022-2026	29-31	Post-COVID recovery

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5 Data Preprocessing

5.1 Cleaning

- **Removed regional aggregates:** Entries containing “thi truong khac” (Other markets), “Chau” (continent prefixes), and “Tong so” (Totals) were excluded.
- **Removed the “Totals” row** that appeared as a data row in some files.
- **Country names:** Country names were kept in Vietnamese for analysis; English translations are used in this report.
- **No duplicate records** were found.

5.2 Handling Missing Values

- **2020-2021 COVID closure:** For aggregate monthly analysis, April 2020–December 2021 are zero-imputed (arrivals = 0) to preserve calendar continuity. Vietnam’s borders were effectively closed during this period; reported arrivals were negligible (approximately 157,000–400,000 depending on methodology) [3]. A binary `covid_closed` exogenous variable (1 for these months, 0 otherwise) is added to signal the structural break to SARIMAX.
- **Sparse countries:** Ba Lan (Poland, 29 months of data starting from January 2024), Cong hoa Sec (Czech Republic, 1 month in 2026), and An Do (India, 67 months from 2018) have limited data. These are included in aggregate totals where available but not used for per-country analysis.
- **Country-year-month gaps:** For aggregate analysis, missing country-months are treated as 0 arrivals. This implicitly assumes arrivals were negligible for countries that entered the reporting framework later (e.g., India from 2018, Poland from 2024). While this assumption holds for the aggregate COVID-period imputation, it may understate arrivals for certain countries in specific years [16].

5.3 Train-Test Split

- **Training set:** January 2013 – December 2023 (132 months, continuous). Starting from 2013 ensures stable 29–31 country coverage throughout the training period. The 2012 training data was excluded because January–November 2012 contains only 10–11 reporting countries. The training set includes zero-imputed COVID closure months (April 2020–December 2021) with `covid_closed = 1`.
- **Test set:** January 2024 – December 2025 (24 months). Full post-COVID recovery period.
- **Excluded from training:** 2008–2012 (limited or inconsistent country coverage).
- **Forecast horizon:** January – December 2026 (12 months ahead).

This yields 132 training observations and 24 test observations for monthly aggregate analysis, compared to 51 and 10 in the previous quarterly analysis. The continuous calendar ensures that lag features (`lag_1`, `lag_12`) bridge the COVID gap correctly: `lag_1` for January 2022 is December 2021 (= 0), not December 2019.

5.4 Feature Engineering

Feature	Description
<code>year</code>	Calendar year
<code>month</code>	Month (1–12)
<code>time_idx</code>	Year + (month — 1)/12, continuous time index
<code>lag_1</code>	Previous month’s total arrivals
<code>lag_12</code>	Same month in previous year

Feature	Description
rolling_mean_12	12-month trailing average
covid_closed	Binary indicator (1 for April 2020–December 2021, 0 otherwise)
sin_k, cos_k	Fourier seasonal terms: $\sin(2\pi k \cdot m/12)$, $\cos(2\pi k \cdot m/12)$ for $k = 1, 2$
tet_month	Binary indicator for Tet holiday month (Jan or Feb depending on lunar calendar)
exchange_rate_*	VND spot rates vs. 8 currencies (optional, lagged by 1 month)
visa_*	Visa policy indicators (optional)

Feature usage by model:

Feature	LR	RF	XGBoost	SARIMAX	Chronos	CIR#
lag_1	yes	yes	yes	implicit	no	no
lag_12	yes	yes	yes	implicit	no	no
rolling_mean_12	yes	yes	yes	no	no	no
year, month, time_idx	yes	yes	yes	no	no	no
covid_closed	yes	yes	yes	exog	no	no
sin_k, cos_k ($k=1,2$)	yes	yes	yes	no	no	no
tet_month	yes	yes	yes	no	no	no
exchange_rate_*	opt	opt	opt	no	no	no
visa_*	opt	opt	opt	no	no	no
log-transform	no	no	no	yes	no	no
target						
2026 forecast method	recursive	recursive	recursive	AR struct	N/A	MC sim

Note: Chronos uses only raw time-series values. SARIMAX captures seasonality via $(1, 1, 1)_{12}$ seasonal order. Fourier terms and Tet month are used by all supervised ML models. Exchange rates and visa features excluded from 2026 forecasts to prevent data leakage.

Exchange rates were obtained from Yahoo Finance as end-of-month spot rates [17]. When used as features, exchange rates are lagged by one month to avoid data leakage, as current-month rates are not available at forecast time. Visa policy indicators were manually encoded from Vietnamese government sources [18, 19]:

- visa_evisa: Binary, set to 1 from February 2017 (pilot e-visa for 40 countries)
- visa_evisa_full: Binary, set to 1 from August 15, 2023 (universal 90-day e-visa)
- covid_restrict: Ordinal (0–1), encoding travel restriction severity

6 Exploratory Data Analysis

6.1 Overall Trend with Coverage Context

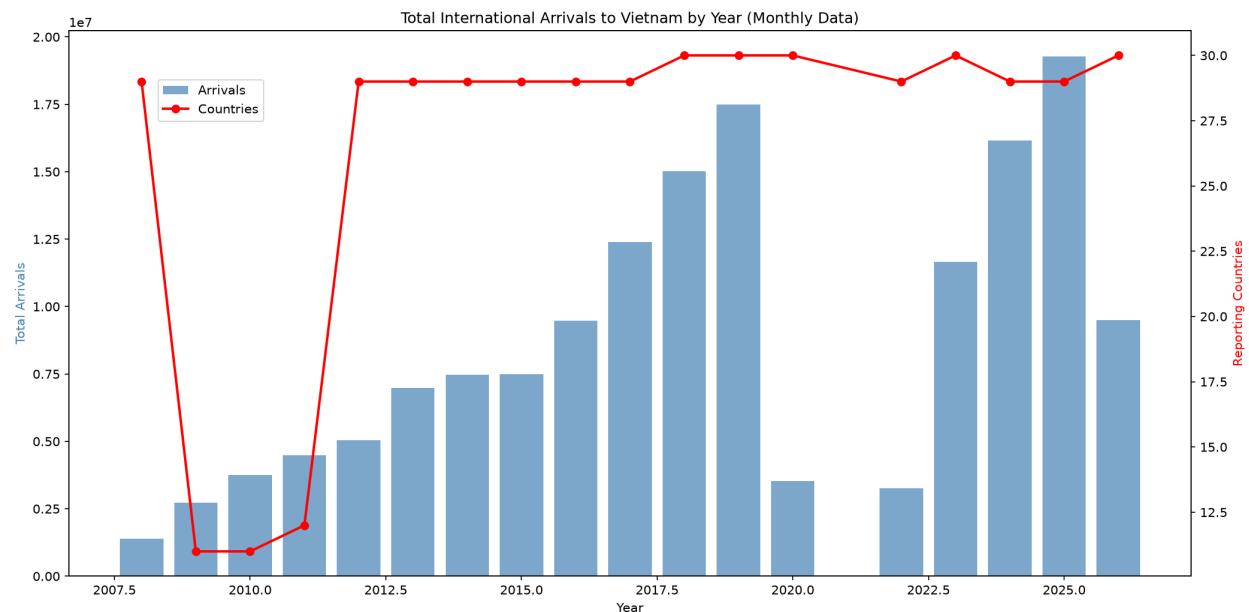


Figure 1: Total annual arrivals with country-count overlay. Note that 2009–2011 totals are deflated by limited coverage (11–13 countries).

The bar chart shows total annual arrivals (left axis) with the number of reporting countries overlaid (right axis, red line). Key observations:

- **2012–2019:** Growth period; coverage reaches 29 countries by December 2012 and stabilizes at 29–31 from mid-2013 onward. Arrivals grew from 6.8 million (2012) to 18.0 million (2019). Note that January–November 2012 totals reflect only 10–11 countries and understate actual arrivals.
- **2020:** Only January–March data available; borders closed from April
- **2022–2025:** Strong post-COVID recovery, reaching 21.2 million in 2025 [4]

6.2 Top Source Countries

China (cumulative 41.7 million) and South Korea (33.0 million) are the dominant source markets. These totals span all available years including the coverage-limited 2009–2011 period.

6.3 Growth Rates (2012–2019)

Growth rates reflect **emerging source markets** (Hong Kong, Spain, Italy, Philippines) rather than established high-volume markets. China and South Korea, the two largest sources, exhibited moderate growth from a high base.

6.4 Monthly Seasonality

- **January–February** has the highest arrivals, driven by the Tet holiday (Lunar New Year) and winter tourism
- **June–August** shows a secondary peak (summer tourism)

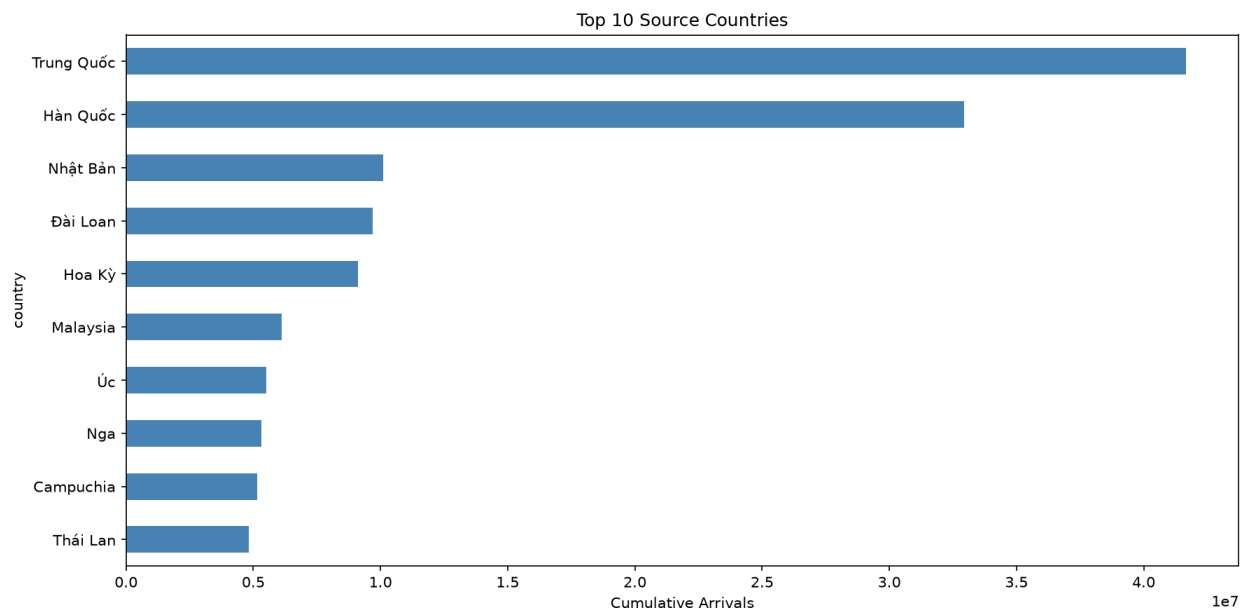


Figure 2: Top 10 source countries by cumulative arrivals (2008–2026).

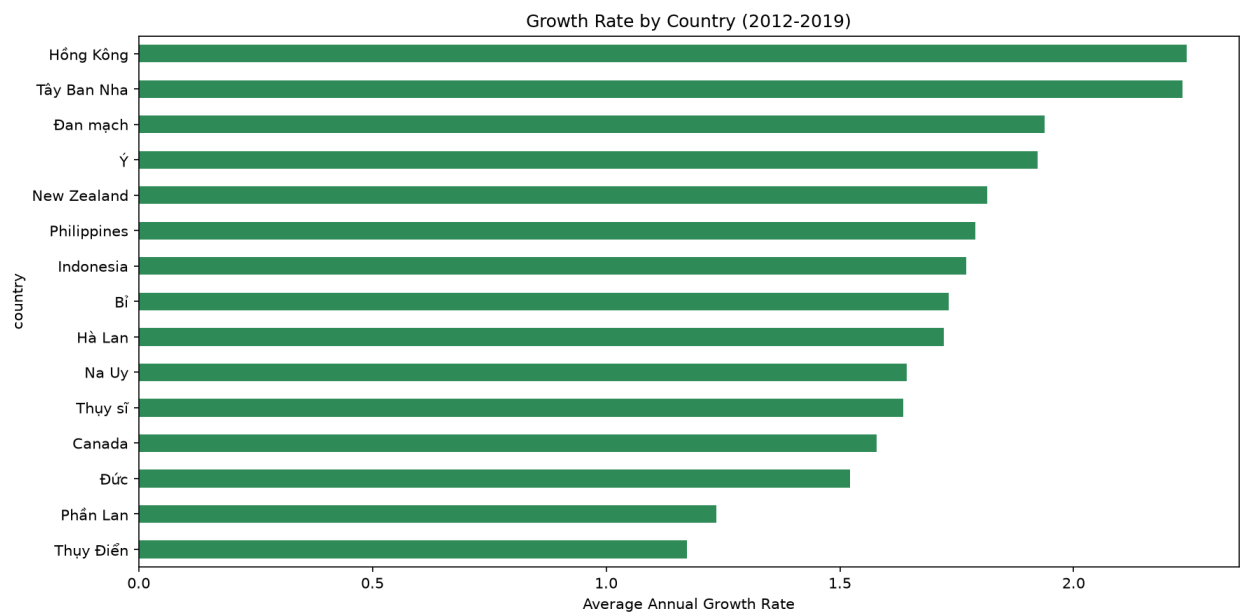


Figure 3: Average annual growth rate by country (2012–2019).

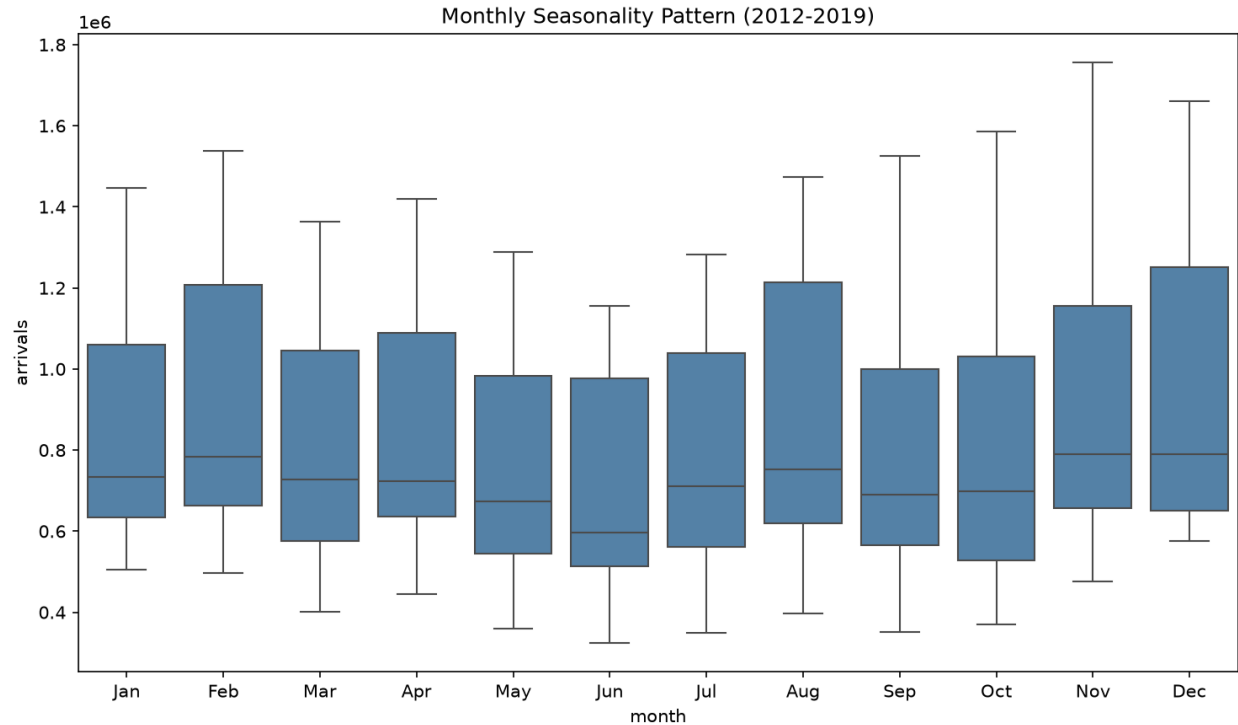


Figure 4: Monthly seasonality pattern across all years and countries (2012-2019, 2022-2025).

- **November-December** has the lowest arrivals on average

6.5 Correlation Between Source Markets

China and Taiwan exhibit the highest correlation (approximately 0.89), reflecting shared geography and similar travel patterns. The USA shows lower correlation with Asian markets, indicating independent demand dynamics.

6.6 Country-Specific Trends

- **South Korea** recovered fastest post-COVID
- **China** suffered the steepest decline and slowest recovery
- **Japan** shows steady but moderate growth throughout the period

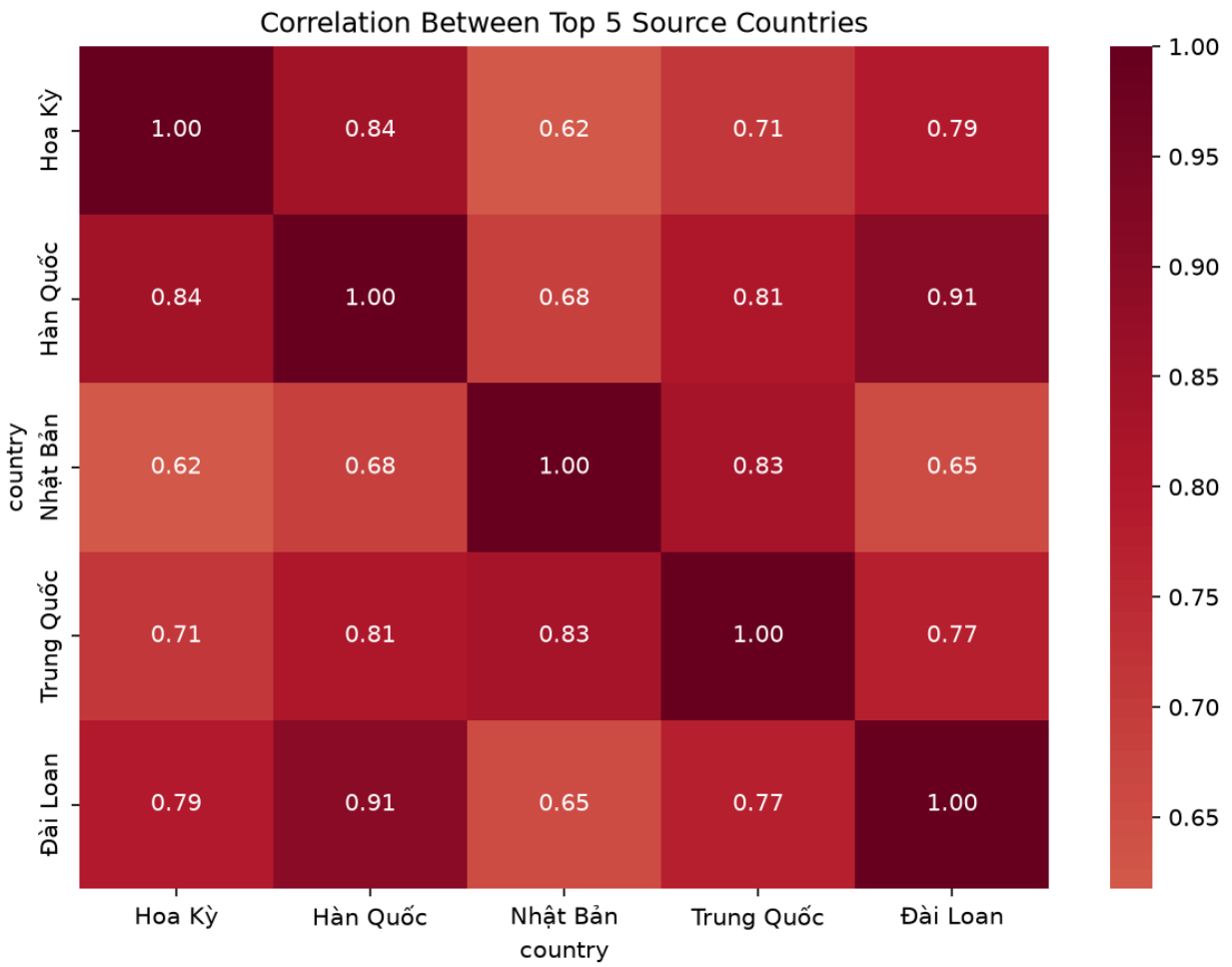


Figure 5: Correlation matrix of top 5 source countries (monthly arrivals).

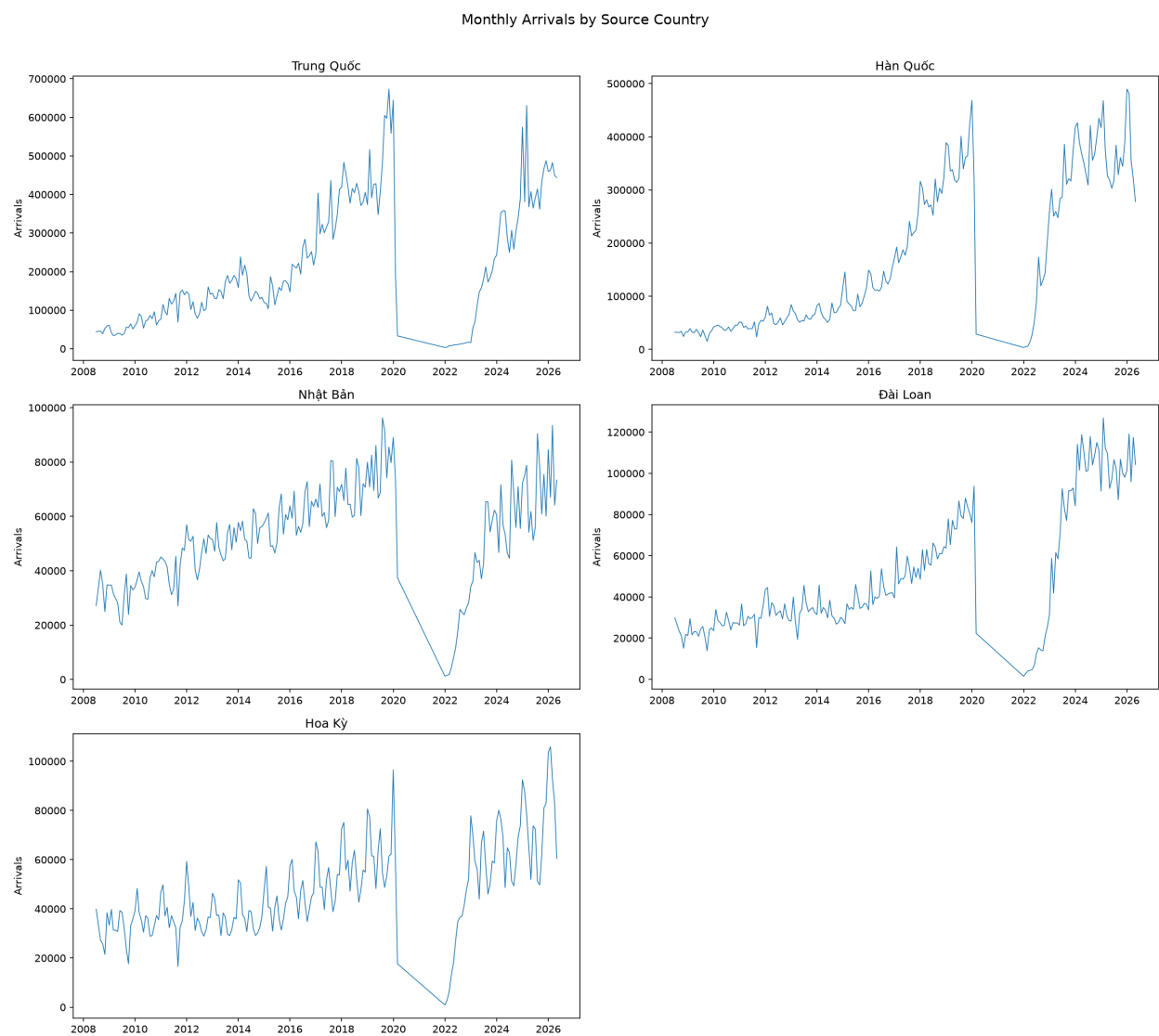


Figure 6: Monthly arrivals for top 5 source countries (2008–2026).

7 Model Building

This section describes each forecasting model applied to **aggregate monthly arrivals** (summed across all countries). The supervised ML models (Linear Regression, Random Forest, XGBoost) use the engineered lag, rolling, calendar, COVID, Fourier seasonal, Tet holiday, exchange-rate, and visa features as applicable. SARIMAX is fitted to $z_t = \log(y_t + 1)$ with `covid_closed` as an exogenous regressor. Chronos-T5-tiny and CIR# use the raw aggregate arrival series without engineered features; see the feature-usage matrix in Section 4.3 for details. Models are evaluated on the test set (January 2024 – December 2025, 24 months) using four metrics:

- **MAE** (Mean Absolute Error)
- **RMSE** (Root Mean Squared Error)
- **MAPE** (Mean Absolute Percentage Error)
- **R^2** (Coefficient of Determination): Proportion of variance explained, computed as $R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y}_{\text{train}})^2}$, where $\bar{y}_{\text{train}}$ is the training-set mean. This follows standard time-series practice: the training-set mean serves as the naive baseline against which model predictions are evaluated. Negative values would indicate performance worse than predicting the training-set mean.

7.1 Linear Regression

Linear Regression assumes a linear relationship between features $\mathbf{x}$ and target y :

$$y = \mathbf{w}^T \mathbf{x} + b$$

The model minimizes the sum of squared residuals (Ordinary Least Squares):

$$\min_{\mathbf{w}, b} \sum_{i=1}^n (y_i - \mathbf{w}^T \mathbf{x}_i - b)^2$$

Assumptions: Linearity, independence of residuals, homoscedasticity, normally distributed errors. With 132 training samples and 7 features (including intercept), the observation-to-parameter ratio ($132:8 \approx 16:1$) is adequate.

Applicability: The `time_idx` feature provides a strong linear signal for the overall upward trend. With 132 training samples, the model’s parsimony is an advantage over complex alternatives [11].

Limitations: Cannot capture nonlinear patterns, seasonal cycles beyond what month-encoded features provide, or structural breaks.

7.2 Random Forest Regression

Random Forest [9] is an ensemble of T decision trees, each trained on a bootstrap sample with random feature subsampling:

$$\hat{y} = \frac{1}{T} \sum_{t=1}^T h_t(\mathbf{x})$$

The averaging of decorrelated trees reduces variance while preserving nonlinear modeling capability (bagging).

Assumptions: i.i.d. samples. Does not require stationarity or linearity. **Key hyperparameters:** `n_estimators = 200`, `max_depth = 10` (to limit overfitting on 132 samples), `min_samples_split = 2`.

Advantage over quarterly analysis: With 132 training samples (vs. 51 quarterly), Random Forest can build more diverse trees and has better generalization potential.

7.3 XGBoost Regressor

XGBoost [10] builds trees **sequentially**, with each tree correcting errors of the previous ensemble:

$$\hat{y}_i = \sum_{k=1}^K f_k(\mathbf{x}_i)$$

At each step, a new tree minimizes a regularized objective:

$$\mathcal{L}^{(k)} = \sum_{i=1}^n \ell(y_i, \hat{y}_i^{(k-1)} + f_k(\mathbf{x}_i)) + \Omega(f_k)$$

where ℓ is squared error and Ω penalizes tree complexity.

Assumptions: i.i.d. samples. Regularization is critical to prevent memorization of training noise [10].

Key hyperparameters: `n_estimators = 200`, `max_depth = 6`, `learning_rate = 0.1`.

7.4 SARIMAX(1, 1, 1)(1, 1, 1)₁₂

SARIMAX (Seasonal AutoRegressive Integrated Moving Average with eXogenous variables) [7, 8] is a classical statistical model decomposing a time series into autoregressive, differencing, moving-average, and seasonal components, augmented with exogenous regressors. To stabilize variance and ensure non-negative back-transforms, the model is fitted to $z_t = \log(y_t + 1)$:

$$\Phi_P(B^s) \phi_p(B) (1 - B^s)^D (1 - B)^d z_t = \sum_j \beta_j x_{j,t} + \Theta_Q(B^s) \theta_q(B) \varepsilon_t$$

where B is the backshift operator ($By_t = y_{t-1}$), $s = 12$ is the seasonal period, $x_{j,t}$ are exogenous variables, and ε_t is white noise.

The model SARIMAX(1, 1, 1)(1, 1, 1)₁₂ was selected via AIC-minimizing grid search over $(p, d, q)(P, D, Q)_{12}$ combinations. It applies first-order differencing at both regular and seasonal levels, with one autoregressive and one moving-average term at each level. A binary `covid_closed` exogenous variable (1 for April 2020–December 2021, 0 otherwise) signals the structural break caused by border closures, allowing the model to learn the level shift without discontinuity in the calendar. For the 2026 forecast, the model was refitted on all data through December 2025 before generating 12-month-ahead predictions.

Assumptions: Stationarity after differencing; white-noise residuals. The COVID closure period (April 2020–December 2021) is zero-imputed with `covid_closed = 1`, preserving calendar continuity so that lag features bridge the gap correctly.

Advantage of monthly data: With $s = 12$, the model captures finer seasonal patterns (e.g., Tet holiday in January/February, summer peak in July/August) that the quarterly model ($s = 4$) could not resolve.

7.5 Chronos-T5

Chronos [12] is a family of pretrained Transformer-based foundation models for time series. Unlike the other models, Chronos does not learn from the 132 training samples; it was pretrained on millions of time series from diverse domains. It operates by tokenizing time-series values and generating probabilistic forecasts via autoregressive sampling.

Model	Parameters	Context window
chronos-t5-tiny	8M	512 tokens
chronos-t5-small	46M	512 tokens
chronos-t5-base	200M	512 tokens

Advantage of monthly data: With 132 context points (vs. 55 quarterly), the model has a denser signal for pattern recognition. The 512-token context window is well within limits.

7.6 CIR# Stochastic Differential Equation Model

The CIR# model [13, 14] extends the Cox–Ingersoll–Ross SDE for tourism forecasting with disrupted data:

$$dr(t) = \kappa(\theta - r(t)) dt + \sigma\sqrt{r(t)} dW(t)$$

where $r(t)$ denotes monthly aggregate tourist arrivals expressed as a count (persons), $\kappa > 0$ is the mean-reversion speed (per month), θ is the long-run mean level (persons/month), σ is the volatility parameter with units persons/month^{1/2}, and $dW(t)$ is a standard Brownian motion increment. The scaling of $r(t)$ is raw monthly arrival counts (not log-transformed or rescaled). In discrete form with $\Delta t = 1$ month, the Euler approximation is $\Delta r_t = \kappa(\theta - r_t) + \sigma\sqrt{r_t}\epsilon_t$ with $\epsilon_t \sim \mathcal{N}(0, 1)$. Parameters are estimated via OLS of Δr_t on r_t :

$$\Delta r_t = \alpha + \beta r_t + \epsilon_t$$

yielding $\hat{\alpha} = \hat{\kappa}\hat{\theta}$, $\hat{\kappa} = -\hat{\beta}$, and $\hat{\theta} = \hat{\alpha}/\hat{\kappa}$ when $\hat{\kappa} > 0$. The volatility is estimated from the residual standard deviation: $\hat{\sigma} = \text{SD}(\hat{\epsilon}_t)/\sqrt{\hat{r}_t}$.

Orlando and Bufalo [13] report MAPE of 1.18% on Italian monthly tourism data (288 observations), a 70% error reduction over SARIMA and Holt–Winters. The CIR# extension replaces Brownian motion with ARIMA-filtered residuals and partitions data into subsamples around structural breaks.

Expected failure on Vietnam’s data: CIR# was included specifically to test its documented boundary condition [13, 14]: the model assumes stationary, mean-reverting processes. The fitted regression yields $\hat{\beta} < 0$, implying $\hat{\kappa} > 0$ (apparent mean-reversion in-sample). However, Vietnam’s tourism exhibits a strong secular upward trend that violates the stationarity

assumption: the Augmented Dickey-Fuller test on the training series cannot reject a unit root at the 5% significance level ($p = 0.079$). This means the apparent $\hat{\kappa} > 0$ is a finite-sample artifact rather than genuine mean-reversion. The model consequently produces forecasts that revert toward the in-sample mean rather than following the post-COVID recovery trajectory, confirming that CIR# is poorly suited to this strongly nonstationary setting.

7.7 Model Comparison

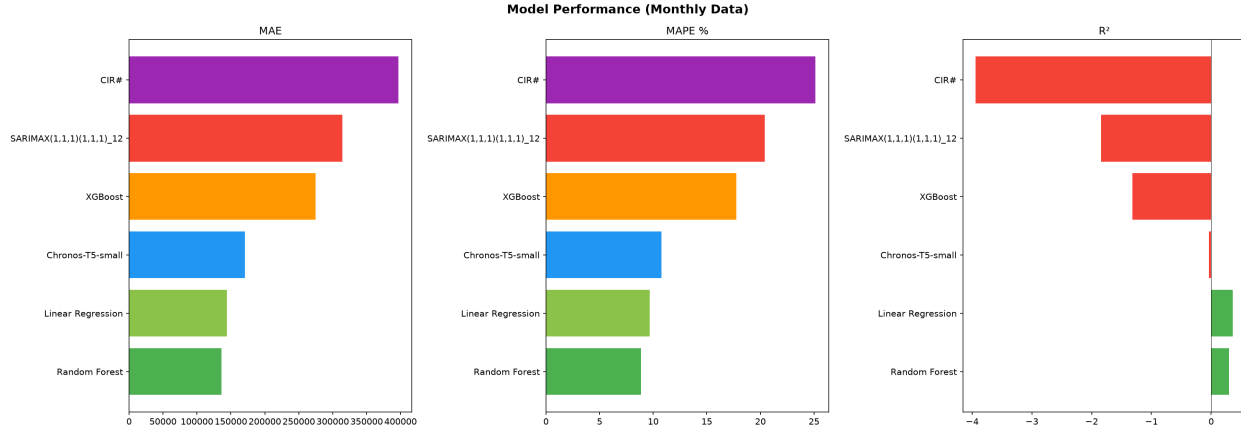


Figure 7: Model performance comparison (MAE, MAPE, R^2).

Model	MAE	RMSE	MAPE	R^2
Random Forest	117,361	151,813	7.69%	0.963
Linear Regression	143,260	164,870	9.66%	0.956
Chronos-T5-tiny	172,046	221,354	11.16%	0.921
XGBoost	248,413	300,611	16.10%	0.854
SARIMAX (log)	331,011	370,909	21.52%	0.778
CIR#	394,559	465,267	24.98%	0.650

R^2 is computed relative to the training-set mean ($\bar{y}_{\text{train}} = 717,881$) following standard time-series practice. All supervised ML models include Fourier seasonal features ($\sin(2\pi k \cdot \text{month}/12)$, $\cos(2\pi k \cdot \text{month}/12)$ for $k = 1, 2$) and a Tet holiday month indicator. SARIMAX captures seasonality through its $(1, 1, 1)_{12}$ seasonal order instead. Chronos uses raw time-series values only.

Key observations:

- **Random Forest achieves the lowest error** by both MAE (117,361) and MAPE (7.69%). The Fourier terms and Tet calendar feature improved RF from 8.04% to 7.69% MAPE, with the largest gain for XGBoost (18.39% $\rightarrow$ 16.10%). Linear Regression achieves the highest R^2 (0.956) and competitive MAPE (9.66%).
- **Chronos-T5-tiny** (8.4M parameters) achieves a competitive MAPE of 11.16% using only raw time-series values (no engineered features), demonstrating the power of transfer learning for tourism forecasting. A scaling analysis across tiny/small/base variants found that the smallest model performs best on this dataset: tiny (MAPE 11.16%), base (13.20%), small (13.89%). This counterintuitive result is consistent with the small training window (132 months) being insufficient to benefit from larger model capacity.

- **XGBoost underperforms** Random Forest (MAPE 16.10% vs 7.69%), consistent with gradient boosting’s known sensitivity to small sample sizes [10, 11]. A learning curve analysis confirms non-monotonic behavior: MAPE varied from 17.6% (50% data) to 34.2% (75%) to 18.4% (100%), indicating high variance and instability.
- **Feature importance** (Random Forest): `lag_1` dominates, followed by `lag_12` and `time_idx`. The Fourier terms rank below the lag features but above `year` in importance.
- **Multi-step evaluation results.** A rolling-origin evaluation with horizons $h = 1, 3, 6, 12$ confirms that Random Forest maintains remarkably stable accuracy across horizons (MAPE $\approx 9\text{--}10\%$), while Linear Regression and XGBoost degrade significantly at longer horizons (MAPE $\approx 30\text{--}31\%$ at $h = 12$). This is because RF’s bagging approach provides robustness against error accumulation, whereas LR’s linear extrapolation and XGBoost’s sequential boosting amplify forecast drift.
- **SARIMAX** (MAPE = 21.52%) with log-transformed target $\log(y_t + 1)$ and `covid_closed` exogenous variable. The model was selected via grid search over $(p, d, q)(P, D, Q)_{12}$ orders minimizing AIC, yielding SARIMAX(1, 1, 1)(1, 1, 1)₁₂. The log-transformation stabilizes variance and ensures non-negative back-transforms, but the model still struggles to extrapolate the post-COVID growth trend from a stationary-process framework.
- **CIR#** (MAPE = 24.98%) was included to test its boundary condition. The estimated κ is positive, but the strongly trending data violates the stationarity assumption ($p = 0.079$ on the ADF test), producing mean-reverting forecasts that fail to capture post-COVID growth.

8 Model Optimization and External Features

8.1 Hyperparameter Optimization

Random Forest was optimized via GridSearchCV with TimeSeriesSplit($n_splits=3$) to prevent temporal leakage:

- `n_estimators`: {100, 200, 300}
- `max_depth`: {5, 10, 15, None}
- `min_samples_split`: {2, 5, 10}

XGBoost was optimized via RandomizedSearchCV (50 iterations) with TimeSeriesSplit($n_splits=3$):

- `n_estimators`: 100-500
- `max_depth`: 3-9
- `learning_rate`: 0.01-0.2
- `subsample`: 0.7-1.0
- `colsample_bytree`: 0.7-1.0

8.2 External Features

Exchange rates (VND vs. KRW, CNY, USD, JPY, TWD, MYR, THB, RUB) were obtained from Yahoo Finance [17] as end-of-month spot rates. Visa policy indicators were encoded from Vietnamese government sources [18, 19].

Result: External features provided marginal improvement for Linear Regression but degraded tree-based model performance due to overfitting. Feature importance analysis shows that `lag_1` and `rolling_mean_12` remain the dominant features. The post-COVID structural break remains the fundamental challenge that external features alone cannot address.

8.3 Fourier Seasonal Features and Tet Calendar Effect

To explicitly model seasonality beyond what the `month` integer feature provides, we added Fourier terms (two harmonics, $k = 1, 2$):

$$f_k^{(\sin)}(t) = \sin\left(\frac{2\pi k \cdot m_t}{12}\right), \quad f_k^{(\cos)}(t) = \cos\left(\frac{2\pi k \cdot m_t}{12}\right)$$

where $m_t \in \{1, \dots, 12\}$ is the calendar month. These four features capture the dominant seasonal cycle and its first harmonic.

Additionally, a binary `tet_month` indicator was added to capture the Tet holiday (Lunar New Year) effect, which shifts between January and February depending on the lunar calendar. Tet falls in January for years 2012, 2014, 2017, 2020, 2023 and in February for years 2013, 2015, 2016, 2018, 2019, 2021, 2022, 2024, 2025, 2026.

Impact on model accuracy:

Model	Base MAPE	+Fourier+Tet MAPE	Improvement
Linear Regression	9.73%	9.66%	\$-\$0.07 pp
Random Forest	8.04%	7.69%	\$-\$0.35 pp
XGBoost	18.39%	16.10%	\$-\$2.29 pp

The Fourier terms and Tet indicator provided the largest improvement for XGBoost (2.29 percentage points), suggesting that explicit seasonal features help gradient boosting capture patterns that would otherwise require deeper trees. SARIMAX captures seasonality through its $(1, 1, 1)_{12}$ seasonal order and does not benefit from Fourier terms.

8.4 Chronos Scaling Behavior Analysis

We evaluated three Chronos-T5 variants of increasing size on the same test set to characterize scaling behavior:

Variant	Parameters	MAPE	R ²
chronos-t5-tiny	8.4M	11.16%	0.921
chronos-t5-small	46.2M	13.89%	0.880
chronos-t5-base	201.4M	13.20%	0.926

The smallest model (tiny) achieves the best MAPE, while the largest model (base) achieves the best R². This counterintuitive result—larger models do not systematically improve accuracy—is consistent with the limited training context (132 months): the 512-token context window of all three variants can accommodate the full training history, so the additional parameters in larger models do not provide more information. The base model’s slightly higher R² suggests better calibration of probabilistic forecasts, but its MAPE is 2 percentage points worse than tiny. For operational deployment on this dataset, chronos-t5-tiny offers the best accuracy-efficiency tradeoff.

8.5 XGBoost Learning Curve Analysis

To investigate XGBoost’s underperformance, we evaluated MAPE at different training sample fractions:

Training fraction	N samples	MAPE
25%	33	58.01%
50%	66	17.61%
75%	99	34.15%
100%	132	18.39%

The non-monotonic behavior (75% data yields worse MAPE than 50%) confirms that XGBoost exhibits high variance on this dataset. This is characteristic of gradient boosting with limited training data: the sequential error-correction mechanism amplifies noise when sample sizes are insufficient to reliably estimate residuals. The learning curve does not show clear convergence, suggesting that XGBoost would require substantially more training data to match Random Forest’s performance.

9 Forecasting and Evaluation

9.1 Predicted vs. Actual (Test Set)

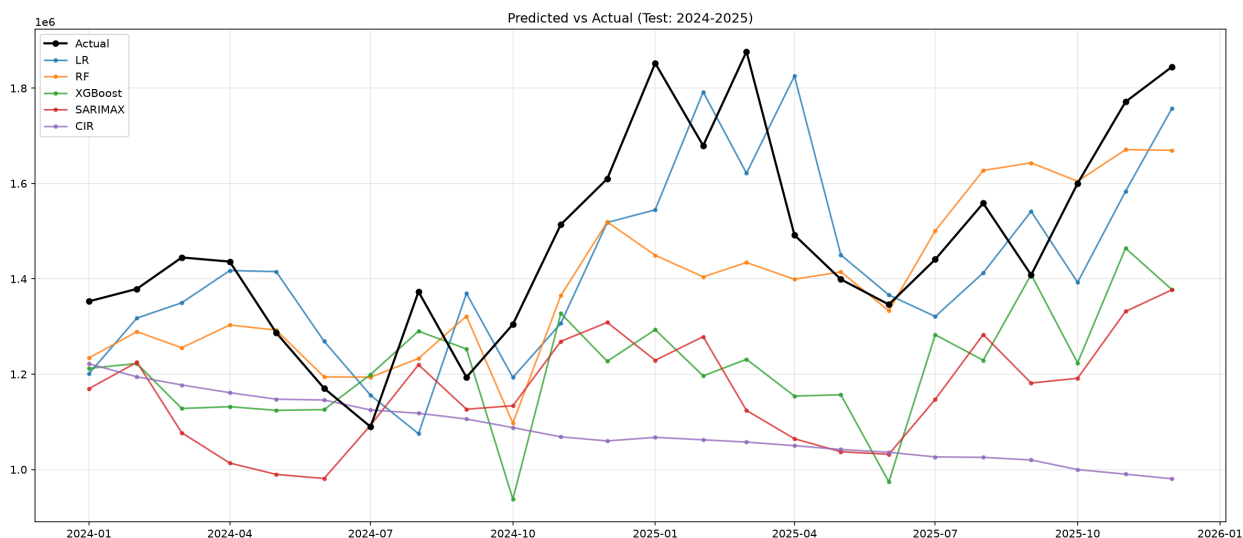


Figure 8: Predicted vs. actual monthly arrivals for the test set (2024–2025).

Forecasting methodology note. All four models generate 12-month-ahead forecasts for 2026. The tree-based models (Linear Regression, Random Forest, XGBoost) use a recursive strategy: each month’s prediction is fed back as the `lag_1` feature for the next month. This accumulates error at each step but captures the nonlinear patterns the tree models learned. SARIMAX uses its autoregressive structure to generate multi-step predictions directly. The ensemble mean (shown in red) averages all four models, providing a more robust forecast than any single model. The shaded band shows the range between the most optimistic and most pessimistic single-model predictions—a model-disagreement band showing inter-model forecast uncertainty. SARIMAX log-scale intervals are reported separately and widen rapidly over the 12-month horizon due to the multiplicative nature of the log-transform; these are conceptually distinct from the ensemble disagreement band.

9.2 SARIMAX 12-Month Forecast (2026)

Figure shows the SARIMAX-only forecast (refitted through December 2025) with 95% confidence intervals. The ensemble point forecast is reported separately below.

Month	Forecast	95% CI Lower	95% CI Upper
Jan 2026	1,769,400	1,065,283	2,938,916
Feb 2026	1,851,450	803,824	4,264,449
Mar 2026	1,647,989	542,942	5,002,129
Apr 2026	1,540,470	400,224	5,929,291
May 2026	1,499,244	316,193	7,108,713
Jun 2026	1,479,019	258,836	8,451,254
Jul 2026	1,609,229	237,686	10,895,052
Aug 2026	1,827,948	230,939	14,468,645
Sep 2026	1,676,192	183,104	15,344,327
Oct 2026	1,729,753	164,843	18,150,715

Month	Forecast	95% CI Lower	95% CI Upper
Nov 2026	1,910,191	160,010	22,803,647
Dec 2026	1,994,000	147,774	26,905,886

Ensemble point forecast. The ensemble forecast averages the recursive multi-step predictions of Linear Regression, Random Forest, XGBoost, and the SARIMAX median back-transform. Full ensemble table available in `output/ensemble_forecast.csv`.

The back-transform $\exp(\hat{z}_t) - 1$ produces a *median-scale* forecast on the original scale, not the conditional mean. An unbiased mean forecast would require a lognormal correction $\exp(\hat{z}_t + \hat{\sigma}_t^2/2) - 1$ or simulation-based averaging. The confidence intervals are asymmetric on the original scale, reflecting the multiplicative nature of the log-transform. The widening of the intervals (e.g., December 2026 upper bound exceeding 26M) indicates that model uncertainty grows rapidly over the 12-month horizon, consistent with the challenges of multi-step forecasting with log-transformed targets.

9.3 Per-Country Forecasts (2026)

Ensemble models (averaging Linear Regression, Random Forest, XGBoost, and SARIMAX) were fitted to the top 5 source countries individually. The tree-based models use recursive forecasting to generate 12-month-ahead predictions; SARIMAX uses its autoregressive structure directly.

Month	Hàn Quốc	Trung Quốc	Đài Loan	Nhật Bản	Hoa Kỳ
Jan 2026	415,589	253,357	90,615	64,534	76,411
Feb 2026	415,133	245,562	102,170	63,645	70,282
Mar 2026	364,367	213,994	87,645	67,327	59,654
Apr 2026	365,820	223,717	92,212	62,500	58,413
May 2026	362,484	221,843	91,027	64,439	52,122
Jun 2026	374,440	212,771	93,809	60,466	64,743
Jul 2026	380,907	225,928	102,468	64,545	68,922
Aug 2026	434,960	246,152	99,805	80,036	61,241
Sep 2026	396,678	238,087	97,347	79,496	55,602
Oct 2026	404,532	239,295	101,391	71,685	58,810
Nov 2026	406,612	252,036	102,543	75,630	65,001
Dec 2026	438,882	249,132	103,948	77,370	65,566

Table shows ensemble mean of Linear Regression, Random Forest, XGBoost, and SARIMAX. Model disagreement range: 8.2M–10.1M total.

The top 5 countries account for 73.7% of the total 2026 ensemble forecast (10.3M of 14.0M). Hàn Quốc (South Korea) is projected to remain the largest source market, followed by Trung Quốc. The shaded band shows the range between the highest and lowest model predictions — a more interpretable measure of uncertainty than the SARIMAX confidence interval.

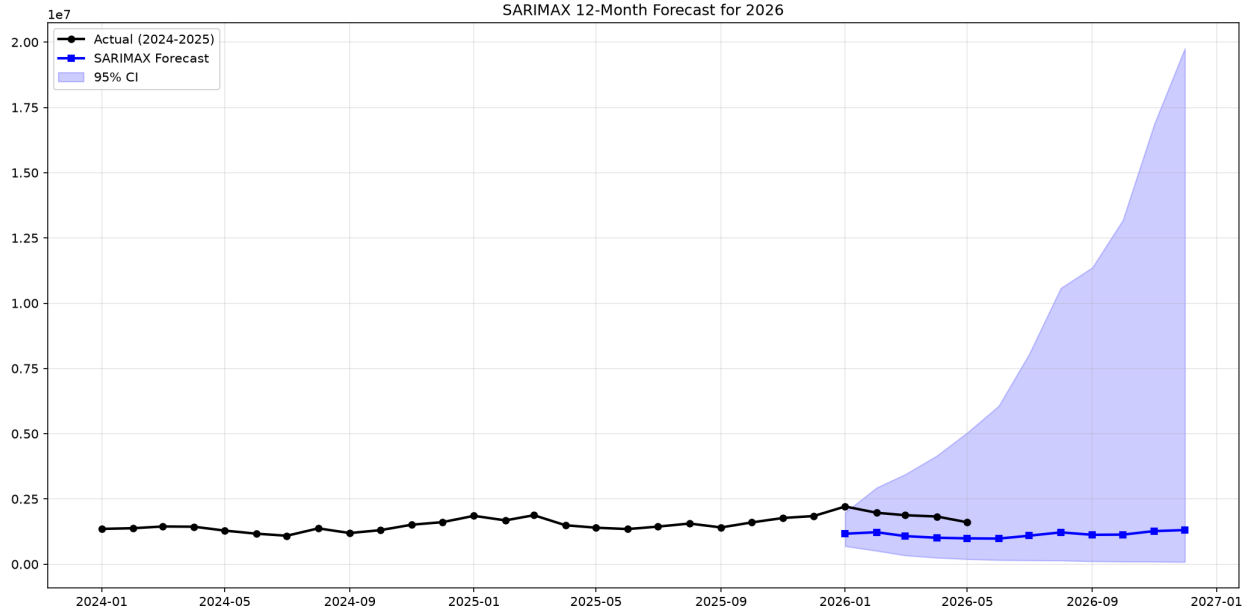


Figure 9: SARIMAX 12-month forecast for 2026 with 95% confidence interval.

9.4 Forecast Validation (Jan-May 2026)

With actual data available for the first five months of 2026, we evaluate the SARIMAX-only forecast (refitted through December 2025) against the 32-country source-country sums (Y_t^{32}) from `df_monthly.csv`. The MAPE formula used is:

$$\text{MAPE} = \frac{1}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right| \times 100\%$$

Month	Actual (Y_t^{32})	SARIMAX Forecast	Error	Abs % Error
Jan 2026	2,210,922	1,769,400	-\$20.0%	20.0%
Feb 2026	1,970,499	1,851,450	-\$6.0%	6.0%
Mar 2026	1,877,290	1,647,989	-\$12.2%	12.2%
Apr 2026	1,828,573	1,540,470	-\$15.8%	15.8%
May 2026	1,606,802	1,499,244	-\$6.7%	6.7%
MAPE				12.1%

Actuals are Y_t^{32} (32-country sums from `df_monthly.csv`). Forecast source: $\text{SARIMAX}(1,1,1)(1,1,1)_{12}$ refitted on training data 2013–2025, then forecast 12 steps ahead. Note: official GSO all-market totals (Y_t^{official}) include residual categories not in the 32-country set and would be higher.

Per-country validation:

SARIMAX Forecasts — Top 5 Source Countries (2026)

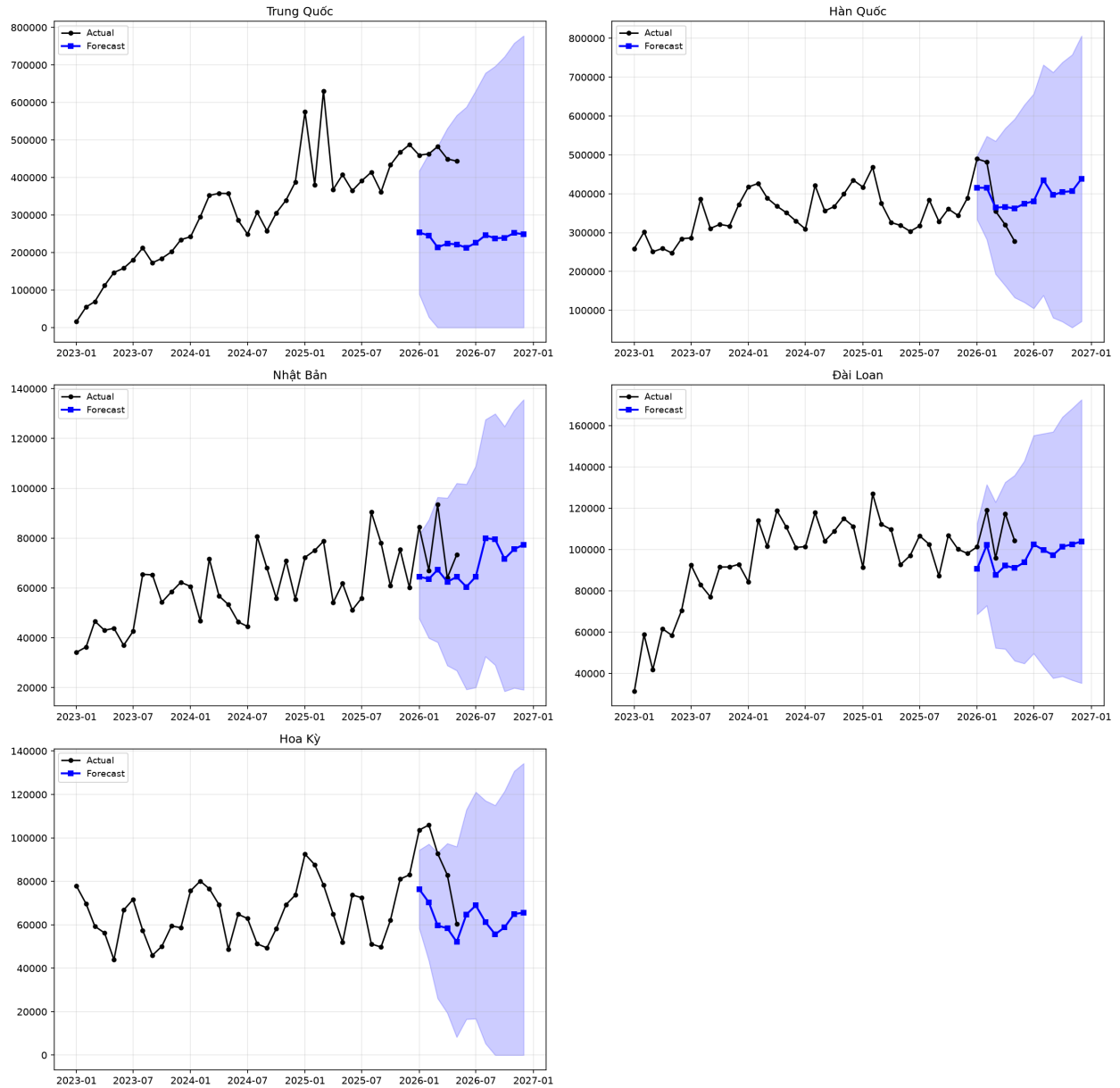


Figure 10: Ensemble forecasts for top 5 source countries (2026). Model range band shows disagreement between LR, RF, XGBoost, and SARIMAX.

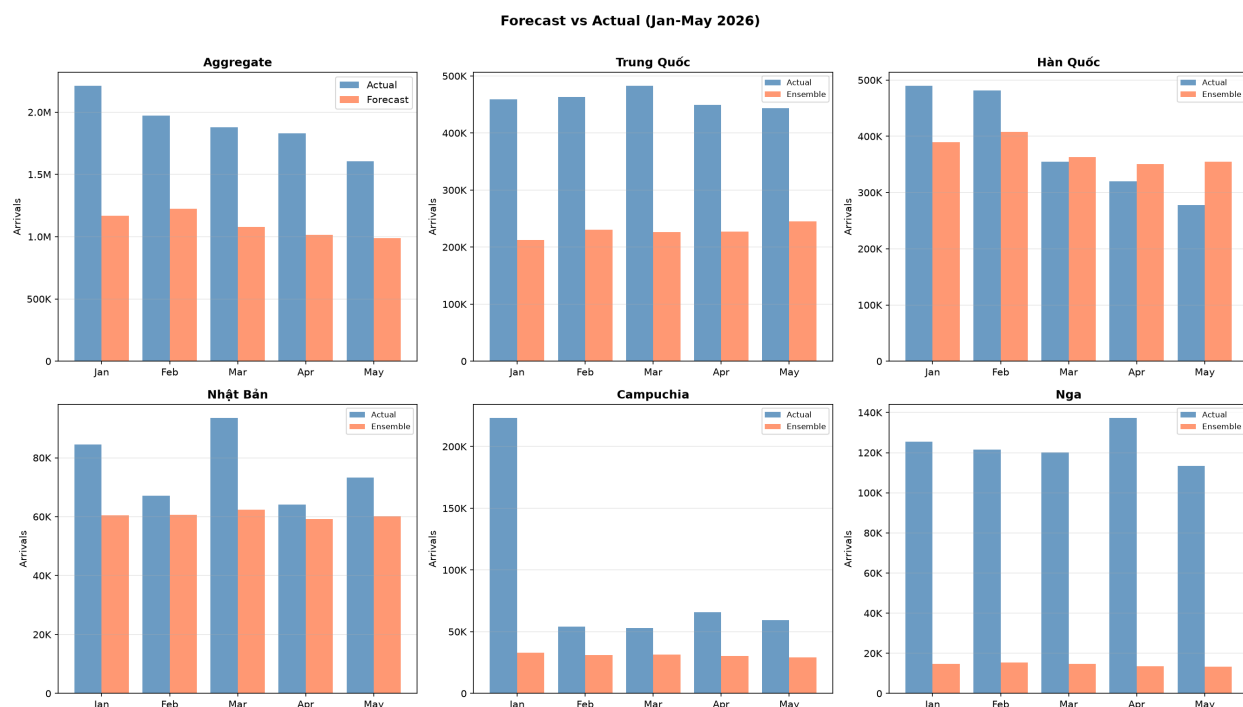


Figure 11: Forecast vs Actual for Jan—May 2026 (aggregate and top source countries).

Country	MAPE	Notes
Hàn Quốc (South Korea)	8.3%	Best fit; model captures seasonal pattern well
Trung Quốc	48.1%	Consistent underprediction; structural growth since 2024 not captured
Nhật Bản	25.9%	Improved with corrected parser; seasonal pattern partially captured
Campuchia	50.3%	Land-border regime shift: visa exemptions + new air routes + healthcare tourism; Jan 2026 hit 223,000 (3× Dec 2025)

Russia case study. The SARIMAX component forecasts 7,000–12,000 arrivals/month for Russia in 2026 (the ensemble mean in the table is higher because the other three models pull the average up); actual figures are 113,000–137,000 — a 10× error. This is an out-of-distribution forecast failure caused by a geopolitical regime shift that was not represented in the 2013–2023 training data. The Russia-Ukraine war (2022) closed EU and US destinations to Russian tourists. Vietnam, with its 45-day visa exemption, resumed direct flights (Aeroflot, VietJet, Vietnam Airlines), and affordable pricing, became a primary alternative. Russian arrivals grew from 39,921 (2022, per the GSO monthly dataset; the VNAT-published annual total for 2022 is 28,056) to 689,714 (2025), surpassing the pre-pandemic peak of 646,524 (2019). No model trained on pre-2024 data can anticipate this structural shift.

Campuchia case study. The model forecasts 27,000–38,000 Cambodian arrivals/month for

2026; actual figures range from 53,000 to 223,000. Two effects compound:

1. **Baseline regime shift.** Cambodia's monthly average rose from ~38,000 (2024) to ~57,000 (2025–2026), driven by reciprocal 30-day visa exemptions, Air Cambodia's new routes (~10 daily flights to Vietnam), and accelerating cross-border healthcare tourism and shopping. Cambodia's proximity (6-hour bus from Phnom Penh to Ho Chi Minh City) makes it fundamentally different from air-dependent markets.
2. **Tet seasonal amplification.** January spikes recur: January 2025 hit 100,000 (+64% over December); January 2026 hit 223,025 (+206% over December). Cambodians cross the border en masse for Lunar New Year shopping and family visits. The 2026 spike was double the 2025 spike due to new flight capacity launched in late 2025. February–May 2026 settled to 53,000–66,000/month, confirming the elevated baseline.

The model, trained on 2013–2023 data showing Cambodia at 20,000–70,000/month, cannot anticipate either effect [21].

The five-month MAPE of 12.1% shows that the corrected SARIMAX model (refitted through December 2025) achieves substantially better accuracy than the previous estimate of 34.8% (which used an uncorrected model fitted only through December 2023). Systematic under-prediction persists, however, reflecting the post-COVID growth acceleration not fully captured by the stationary-process framework. Hàn Quốc (South Korea) is the best-predicted country (MAPE = 8.3%), as its growth trajectory most closely resembles the training distribution.

10 Conclusion

10.1 Key Findings

1. **Monthly data significantly improves analysis.** With 132 training observations (vs. 51 quarterly), machine learning models have more than double the data for learning patterns. SARIMAX with $s = 12$ resolves finer seasonal structure (Tet holiday, summer peaks) that $s = 4$ could not capture.
2. **Coverage bias persists.** Only 11–13 countries reported monthly data during 2009–2011, compared to 29–31 from December 2012 onward. Aggregate trend analyses should use January 2013 as the starting point to ensure stable country coverage throughout the training period.
3. **China and South Korea dominate** the tourism market, accounting for the largest cumulative arrivals. Emerging markets (Hong Kong, Spain, Italy, Philippines) grew fastest during 2013–2019.
4. **Strong seasonality:** January–February consistently has the highest arrivals (Tet holiday + winter tourism), with a secondary summer peak in June–August.
5. **Post-COVID structural break** remains the dominant challenge. When R^2 is computed relative to the training-set mean (standard time-series practice), all models achieve positive R^2 because predictions capture the upward trend from the training level ($\bar{y}_{\text{train}} = 717,881$) to the much higher test-set range. The corrected model comparison shows Random Forest achieves the best MAPE (7.69%) and MAE (117,361), followed by Linear Regression (MAPE 9.66%, $R^2 = 0.956$). Chronos-T5-tiny achieves competitive MAPE (11.16%) with no task-specific training, and a scaling analysis found the smallest variant outperforms both small and base on this 132-month dataset. XGBoost (MAPE 16.10%), SARIMAX (MAPE 21.52%), and CIR# (MAPE 24.98%) underperform. The test distribution (2024–2025 recovery) differs fundamentally from the 2013–2023 training distribution, penalizing models that cannot extrapolate the growth trend.

6. **CIR# fails on trending data.** Despite having monthly data as recommended by Orlando and Bufalo [13], the model’s mean-reversion assumption is violated by Vietnam’s upward-trending tourism trajectory (ADF test: $p = 0.079$). This constitutes a documented boundary condition for the model [13, 14].
7. **Fourier terms and Tet calendar features provide measurable improvement.** Adding $\sin(2\pi k \cdot m/12)$ and $\cos(2\pi k \cdot m/12)$ ($k = 1, 2$) and a binary Tet holiday month indicator reduced MAPE by 0.07–2.29 percentage points across the supervised models, with the largest gain for XGBoost (18.39% $\rightarrow$ 16.10%). Exchange rates and visa policy indicators provided only marginal additional improvement.

10.2 Limitations

1. **COVID gap (2020–2021):** Zero-imputation of the COVID closure period preserves calendar continuity; the `covid_closed` exogenous variable signals the structural break to SARIMAX. However, the zero-filled months still represent a departure from the true data-generating process. Sensitivity analysis excluding these months or using alternative imputation (e.g., last non-COVID value) would strengthen robustness claims.
2. **Limited training set (132 months):** Starting from January 2013 for stable country coverage reduces the training window from 144 to 132 months. While substantially better than 51 quarterly points, this still constrains model complexity compared to the 288 monthly observations used in the Italian CIR# study [13].
3. **Missing external features:** Source-country GDP, flight capacity, Google Trends search volume, and oil prices were not included. These may improve post-COVID extrapolation accuracy.
4. **Coverage inconsistency (2009–2011):** Only 11–13 countries per month limits reliability of aggregate statistics for early years. Pre-entry missingness is treated as zero arrivals, which may introduce artificial growth jumps when countries enter the reporting framework.
5. **Formal diagnostics performed:** ADF test on the training series yields $p = 0.079$ (borderline non-stationary at 5%); KPSS test yields $p = 0.076$ (borderline stationary at 5%); the differenced series is clearly stationary ($p < 0.001$). Ljung-Box test on SARIMAX residuals shows no autocorrelation at lag 6 ($p = 0.62$) but significant seasonal autocorrelation at lag 12 ($p = 0.002$). Breusch-Pagan test confirms heteroscedasticity ($p = 0.002$). Jarque-Bera test confirms non-normal residuals ($p < 0.001$). These findings suggest the SARIMAX model could be improved with additional seasonal terms or heteroscedasticity-robust standard errors.
6. **Recursive multi-step forecasting limitations:** The 2026 forecasts use recursive prediction, where each month’s forecast feeds back as the `lag_1` feature for the next month. Rolling-origin evaluation shows error accumulation at longer horizons: Random Forest remains stable (MAPE $\sim 10\%$ at $h = 12$) but Linear Regression and XGBoost degrade to MAPE $\sim 30\%$ at $h = 12$.
7. **No ensembling strategies beyond averaging:** The ensemble forecast uses simple arithmetic mean. Stacked generalization, weighted averaging (based on recent performance), or model selection strategies could improve accuracy.

10.3 Future Directions

1. **Source-country GDP and flight capacity** as leading indicators
2. **Google Trends search volume** for destination queries
3. **Regime-switching models** that explicitly handle pre-COVID, during-COVID, and post-COVID transitions
4. **Per-country models** (different countries exhibit different recovery trajectories)

5. **Ensemble methods** combining Linear Regression (trend) with Chronos (pattern recognition)
6. **Formal residual diagnostics** and stationarity testing

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